

BRINGING AUTOMATED FAULT DETECTION AND DIAGNOSTICS (FDD) TOOLS FOR HVAC&R INTO THE MAINSTREAM

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ABSTRACT

Heating, ventilation and air-conditioning (HVAC) systems consume over 5 quads of energy annually, representing 30% of energy consumption in the U.S. in commercial buildings. Additionally, commercial refrigeration (R) systems add about 1 quad to commercial buildings energy consumption. Most HVAC systems operate with one or more faults that result in increased energy consumption. Fault detection and diagnostics (FDD) tools have been developed to address this national issue and many tools are commercially available. FDD tools have the potential to save considerable energy for an existing commercial rooftop unit (RTU) and refrigeration systems. These devices can be used for both retro commissioning, and, when faults are addressed, continuous commissioning. However, there appears to be multiple market barriers for this technology. Although there are efforts to develop FDD tool standards, currently there are no standards and methods to define functions, capabilities, accuracy, and reliability of FDD tools in the field. Moreover, most of the commercial FDD tools have not been verified in the field independently. This paper presents a comprehensive approach to bringing HVAC FDD tools into the mainstream. The approach involves demonstrating ten commercially available FDD tools installed at ten different sites, independent testing and evaluation of the FDD tools, communication with various stakeholders, identifying and assessing market barriers, creating a process evaluation methodology, and assisting utility companies in developing incentive programs. The preliminary baseline results from the case study demonstrate how the use of an independent monitoring system (IMS) can be used for ground-truth in evaluating FDD tools in the field.

Keywords: Independent Monitoring System, Heating ventilation and air conditioning, fault detection and diagnostics tools, energy, refrigeration, faults, RTU, Building Management System.

1. INTRODUCTION

Heating, ventilation and air-conditioning (HVAC) systems consume over 5 quads of energy annually, representing 30% of energy consumption in the U.S. in commercial buildings [1]. Packaged rooftop units (RTUs) provide heating and cooling for over 60 percent of the commercial building space (about 90 billion ft²) in the U.S. and they are a significant source of energy consumption and peak demand. It is estimated that 40,000 10-ton RTUs are sold each year in the U.S. There are over 486,000 RTUs in the Northeast region and about 2,700 units sold in 2014 in Connecticut. Another important market segment is commercial refrigeration. It contributes about 1 quad to commercial building energy consumption [2].

Most RTUs have one or more fault types such as low/high refrigerant charge, valve leakage, condenser/evaporator fouling, filter/dryer restriction, economizer, and controls. These types of faults can increase RTU energy consumption. If these faults are detected, diagnosed, and addressed then significant energy could be saved. Automated Fault Detection and Diagnostics (AFDD) tools have the potential to save considerable energy for existing commercial RTUs. These devices, when installed on RTUs and other HVAC systems, can be used for retro commissioning. In addition, when faults are addressed, AFDD tools can be used for continuous commissioning. According to Navigant Consulting [2], national energy savings of 111 TBtu can be saved by employing FDD tools for commercial RTUs. Faults in commercial refrigeration systems not only result in significant energy use, but also can lead to shutdown of equipment that can

cause damage to refrigerated products. Goetzler, et al. [1] indicated that a popular supermarket employed a leakage detection and energy usage monitoring system that reduced electricity use by 23 million kWh per year.

To address this need, countless efforts to develop FDD tools have been ongoing for at least three decades [3–8] and now there are several commercial FDD products marketed by companies. These products include field-portable FDD tools, factory-installed on-board FDD tools, hardware-based retrofit on-board FDD tools, and software-as-a-service (SaaS) FDD tools. However, FDD systems have not yet achieved a high HVAC market penetration. In an online article, “FDD Going Mainstream? Whose Fault is it?,” several issues with FDD implementation are listed, such as: lack of data, rules specific to systems, how to handle the FDD information, using the diagnostic data, prognostics data, and alternative ways to deploy FDD [9]. A key market barrier for this technology appears to be the lack of awareness of FDD products among building owners, who are the majority of potential customers. Most HVAC contractors are not familiar with the latest FDD technologies and HVAC technicians lack knowledge and skills regarding these technologies. Quantifying potential benefits to building owners is difficult. Moreover, many building owners look for short-term return on investment (ROI) for their investments. Many of the hard to quantify FDD benefits, such as reduced HVAC downtime due to early warning and repairs, avoidance of catastrophic failures, and predictable maintenance, will not be so evident to the building owners. FDD technology appears to be like an “unknown saint.” It has promise but is not fully

understood, evidenced, or exploited. To make matters more complicated, many companies package FDD tools with energy-efficiency retrofit kits, and it is not clear to customers which ones to use. Even though there are efforts to develop standards, currently there are no standards and methods to define functions, capabilities, accuracy, and reliability of FDD tools. Many potential customers do not have the tools to evaluate the cost of FDD tools or to assess real value and understand the value proposition. In addition, it is not clear how the process of FDD installation and the communication of faults, severity of faults, and actions occur during use. It is quite clear that addressing these issues and overcoming these barriers are challenging, but necessary. In that regard, there is a continued push for bringing FDD products to the commercial market. Western HVAC Performance Alliance has been working on an FDD Road Map and released a Master List of existing FDD products (over 100) [10]. Some efforts by utilities, federal agencies, and state agencies are underway to bring attention to FDD tools. For instance, California Energy Commission's Title 24-Part 6 requires that economizer FDD tools be installed on air-cooled unitary air conditioning systems over 4.5 tons cooling capacity, with the ability to detect faults.

The uniqueness of this study approach is its comprehensive nature. It includes identification of diverse FDD products, selection of different commercial building types for the field demonstrations, process evaluation, performance verification, determination of technical and economic viability, supporting the development of utility incentives, education, and outreach. Identification of technical and market barriers, and

development of strategies to address them by bringing together all stakeholders are critical aspects of the study. Regarding establishing economic viability of AFDD tools, one of the aims of the study is to determine the energy savings due to AFDD implementation in the field under naturally occurring fault conditions rather than introducing artificial faults. For achieving this, AFDD tools are installed on RTUs in the field and the energy use is monitored during the baseline period. At the end of the baseline period, faults identified by the AFDD tools are verified and any repairs and adjustments needed are implemented as a process of retrofitting. The energy use following the baseline testing is monitored (considered a part of continuous commissioning) and the energy savings due to AFDD tools is calculated considering weather normalization. Since there might be a significant variability in naturally occurring faults for different sites, and there are differences in features of AFDD tools, there is no direct comparison made among different AFDD tools regarding their performance. The AFDD tool capabilities and results are presented independently as individual case studies. This paper presents the first phase of the project from AFDD tool selection through initial baseline testing results. The results of phase II (post-retrofit testing), which will conclude in August 2021, will be presented in a future paper.

2. FDD TOOL AND SITE SELECTION

2.1. FDD Tool Selection

The goal was to select ten different commercially available FDD tools for the field demonstration study; therefore, an FDD tool selection matrix was developed. The matrix included various performance parameters of FDD such as the tool type (whether hardware-based or SaaS-based), type and range of faults detected, capability to detect heating and cooling faults, feature to diagnose and verify faults, limitation on the size of HVAC system, fault communication and frequency, skill level needed to use the tool, FDD tool pricing, etc. FDD vendors received a Request for Information (RFI) that included the questions from the selection matrix. The study received responses from twelve vendors. Based upon the responses, each tool received a normalized score for each performance metric. Table 1 shows the scoring results. Of these twelve FDD tools, one tool is not market ready and one FDD tool vendor dropped out from the study. Therefore, the study considered the remaining ten FDD tools for field-testing and evaluation.

Insert Table 1 Here.

2.2. Site Selection

The project's target was ten sites for the field demonstration of the FDD tools, with the goal of matching one tool per site. The project team developed the following FDD installation criteria: (1) commercial or industrial building type, (2) ease of access to

rooftop units and corresponding ductwork, (3) RTU age, (4) a diverse portfolio, or building set, to demonstrate a variety of cases to potential customers, and (5) the ability to install a varied set of software vs. hardware FDD tools across the ten sites. Table 2 shows all criteria details. Connecticut utility partners, United Illuminating, and Eversource sent a memorandum of understanding (MOU) along with the project scope to several of their customers for the purposes of developing candidate sites. The research team conducted several site visits after receiving MOUs. During these visits, the team validated responses submitted as part of the MOU process and collected additional information for site criteria, such as ease of access to equipment. A set of ten sites were selected from 15 candidate sites and each site was paired with one of the ten FDD tools, as shown in Table 3.

Insert Table 2 Here.

Insert Table 3 Here.

3. FIELD TESTING METHODOLOGY

The purpose of the field demonstration is to study and verify the technical feasibility, economic feasibility, ease of installation, operational impacts, and environmental impacts of FDD through the field installation of FDD tools and independent monitoring.

Section 3.1 describes the background and basis for the evaluation methodology.

Sections 3.2 through 3.5 describe the field-testing methodology.

3.1. BACKGROUND

Braun and Yuill [11,12] developed methods to evaluate and assess the performance of FDD protocols for air conditioning devices (RTUs and split systems) using a fault intensity (FI) and fault impact ratio (FIR). The study demonstrated that their developed FDD evaluator can be used to evaluate the strength and weakness of the FDD protocol by implementing the Refrigerant Charge and Airflow (RCA) diagnostics protocol and normal model approach (developed by a non-faulted set of a data). Five scenarios of test outcomes are presented in the FDD protocol (no response, correct, false alarm, misdiagnosis, and missed detection) for six types of faults (refrigerant charge, low-side and high-side heat transfer faults, liquid line restriction, non-condensable gas in the refrigerant, compressor valve leakage). The authors used the approach of feeding the FDD protocol with a different set of experimental input data under different conditions and then observing the responses. They used the normal model set of data and four versions of the RCA protocol as a case study to demonstrate the use of their developed evaluation methodology. The RCA protocol was found to perform poorly using their evaluation methodology, resulting in up to 51% of faults where no faults were presented, 26 % misdiagnosing faults, and 32 % not detecting faults where the faults were presented.

Wall and Guo [13] studied next-generation automated fault detection and diagnostics (AFDD) tools for commercial building energy efficiency. The results of the study reported benefits and outcomes of AFDD and performance evaluation of AFDD.

The authors studied six different AFDD tools at six different buildings located in Australia for different space types (office, airport, museum, hospital, and laboratory). They connected the AFDD tools to a BMS to obtain real-time data, and the results for each building were reported in terms of energy reduction in annual energy savings, National Australian Built Environment Rating System (NABERS) ratings, improved occupant comfort, avoidable energy, maintenance issues, and site energy intensity. The study also showed the top issues identified in each building.

Dexter and Pakanen [14] studied demonstrating AFDD methods in operating buildings in twelve countries (Belgium, China, Canada, Finland, France, Germany, Japan, Netherlands, Sweden, Switzerland, United Kingdom, and the United States of America) to develop different computer-based demonstration systems, which can interface with the HVAC system. The authors developed twenty-three prototype performance-monitoring tools and three performance validation tools based on artificial intelligence techniques (neural networks, fuzzy logic, expert and rule-based system, case-based reasoning, bond graphs, and qualitative methods). The authors studied twenty-six different FDD tools and approaches for different HVAC systems in different buildings for different space types (factory, hotel, lab, office, research and development center, college, school, and indoor swimming pool). The report defines three main groups of faults (natural fault, artificial fault, simulated fault) that are considered in the study.

Jagpal [15] conducted a new study based on the results obtained in a previous study [14] for demonstrating AFDD methods in operating buildings to illustrate the benefit of a

computer-aided FDD system by working with FDD vendors, contractors, customers, and end users. The report considered three main objectives: (1) assessing the cost-effectiveness of FDD methods, (2) building a prototype computer-aided performance evaluation system that has the ability to detect the performance deficiency and diagnose faults, and (3) investigating the feasibility of developed performance evaluation systems by testing them in operating buildings. The author identified that the availability of measured data and faults that have similar symptoms are the main barriers to identifying the HVAC faults.

Most of the approaches used in the previous study cases and literature are based upon laboratory experiments where authors introduced faults to the system to study fault behavior and to see if the FDD protocol is responding to the fault. However, it is difficult in this study to convince building managers to agree to introduce failures and faults into real-time operating systems currently servicing operating buildings. Therefore, the study team developed the monitoring and validation system and implemented it in parallel with the FDD tool to obtain real data and to evaluate the faults indicated by the FDD products based on FI and FIR of each fault. The study team developed a measurement and verification (M&V) plan based on the rules and techniques illustrated in previous research [11,12,16–18] to show the feasibility of market-available FDD tools by testing them in a set of real-time operating commercial and industrial buildings.

One key objective of the study is to undertake a comprehensive evaluation of FDD tools in the field, which is a complex task that requires examination of the FDD tool performance, cost, ease of implementation, ease of use, data requirements, training requirements, and applicability to the needs of a particular site or customer [19]. A framework proposed by Frank, et al. [20] for FDD performance evaluation was reviewed carefully. The framework is found to be useful for evaluating FDD protocols, but not for evaluating FDD tools in the field under naturally occurring faults, which can vary from no-fault to single faults of varying intensity and frequency, to multiple faults. Therefore, there is no control over some aspects of input samples for the FDD tools. However, the framework principles are used in the FDD tool performance verification. For instance, three categories of faults will be considered – 1) condition-based, 2) outcome-based and 3) behavior-based. Also, ground-truth is employed in the verification of faults by FDD. Ground-truth is defined by a combination of independent monitoring of the RTU with essential sensors and instrumentation, and in-the-field checking and observation of the system, its operation, and controls.

3.2. Independent Monitoring and Validation System

The study team developed a measurement and validation plan based on methodologies developed by Braun and Yuill [11,12] and Mehrabi and Yuill [16–18]. The plan is also based upon the framework developed by Frank et al. [20]. The plan considers faults of an RTU and uses FI and FIR measures to identify them and to calculate their energy

impact. The identification and impact of faults associated with the following subsystems and flows (energy and refrigerant) are considered: (1) refrigerant charge, (2) refrigerant quality, (3) economizer, (4) controls, (5) condenser, (6) evaporator, (7) expansion device, (8) compressor, (9) refrigerant-liquid lines, and (10) sensors as well as the RTU system level effectiveness of (11) energy performance degradation. Based upon possible faults occurring in the subsystems listed above and the fault impact calculations employed by the methodology in Sec. 3.4, the team determined the required components and sensors needed for collecting input data. The Independent Monitoring System (IMS) evaluator tool uses sensors and meters to collect real-time data as shown in Table 4. Table 4 presents the specifications and planned locations of monitoring system sensors and instruments. These sensors are capable of measuring both refrigeration side and air side performance. In addition, the monitoring system is capable of calculating air side and refrigerant-side performance indicators such as energy efficiency ratio (EER) and cooling capacity. Complete specification of all sensors in the FDD evaluator sensor network is shown in the appendix.

Insert Table 4 Here.

A comprehensive IMS evaluator tool comprising a remotely accessible data acquisition system along with essential sensors and instrumentation has been designed and used in the verification process. IMS is literally a laboratory installed in the field on an RTU with over twenty sensors. One-minute interval data is stored and can then be downloaded remotely through a cellular modem. This prevents the project team from having to

return to the site to collect data. A schematic of this Intellilogger data acquisition system is shown in Fig. 1.

Insert Fig. 1 Here.

The location of each sensor was determined based on the required input data collection needed. Based on the initial site visit, the team was able to customize the required sensors and instrumentation according to the RTU characteristics and the location of supply and return ducts for the building. A schematic of an RTU with locations of all sensors is shown in Fig. 2.

Insert Fig. 2 Here

3.3. Post Processing and Fault Verification

A majority of the FDD tools require a building management system (BMS), a building automation system (BAS), or energy management system (EMS) to be installed to read the data points within a RTU. A BMS is an intelligent system that controls and monitors a building's main technical components, such as the HVAC and lighting. Since BMS-based FDD tools can monitor all the RTUs on a building without additional hardware expense, the FDD tool will be setup to monitor all RTUs on a building. However, the independent monitoring system (IMS) is installed only on one RTU for validation. For non-BMS, hardware-based FDD tools, one FDD tool will be installed along with the IMS on one RTU per building. In general, all FDD tools share the same four general installation and use processes: (1) receive and install the tool, (2) verify functionality of

the tool, (3) allow the tool to run over baseline period, (4) review the faults after the baseline period, (5) implement repairs or controls changes to address faults. Many FDD tools require additional tuning processes for more accurately reading and detecting faults.

The team developed an M&V plan using an independent monitoring tool. The plan outlines the steps for the tool pre-installation, installation, data collection, analysis, and post-processing. For pre-installation, the team visited each study site to collect information and field measurements. These included RTU make/model, duct dimensions, determining pressure transducer/temperature sensor locations, and determining RTU accessibility. The team then used the field information to customize each monitoring system to the site. After ordering and receiving the sensors and instrumentations, the team pre-wired and calibrated the monitoring system and verified operation before field installation. A licensed HVAC technician installed all monitoring tools in the field with support from the project team. The HVAC technicians spent 1-2 days installing each monitoring tool in the field, which varied depending upon unit accessibility and a technician's skill and experience level. Once an FDD tool and the IMS were installed, the RTU was monitored for six to eight weeks and the data was collected based upon naturally occurring faults. The research team reviewed and analyzed the real-time data from the IMS weekly.

3.4. Methodology for evaluating FDD tool capabilities

A complete set of fault algorithms has been implemented as part of the verification of FDD tools, based upon previous work by Yuill and Braun [12] and the Pacific Northwest National Laboratory (PNNL) [7]. Additional calculations from the ASHRAE Handbook of Fundamentals [21] have been used during this development, as well as an online thermophysical property tool from NIST [22]. This section provides the methodology for identifying RTU faults using the monitoring system by calculating the FI, as well as examining the impact of these faults on the energy efficiency and energy consumption through the FIR. The methods that will be described are selected based on practicality as well as time constraints of the field study. The FIR can be defined as the ratio between the faulted and non-faulted values of the metric of interest. If the FIR passes a certain threshold determined by the user, then a system is considered “faulted”. The FI can be defined in terms of measurable numeric quantities related to the condition of the fault [12]. This approach can also be used to sort the prevalence of a fault. By setting a higher threshold, fewer alarms are detected. All calculations for this study are for constant-speed equipment rather than variable-speed. The Coefficient of Performance (COP) is calculated based upon real-time input data from the monitoring system for purposes of determining FDD tool performance and calculating energy savings.

Refrigerant charge fault: The maximum performance of an HVAC-R system may occur at lower or higher charge levels than the nominal charge level. Therefore, an accurate

charge level is required for a maximum performance, and the charging table from the manufacturer is not helpful. Braun and Yuill [11] use Eq. (1) to calculate fault intensity for refrigerant charge (FI_{ch}) where m_{act} is actual mass of charge and m_{nom} is nominal correct mass of charge and found in a manufacturer's specification.

$$FI_{ch} = \frac{m_{act} - m_{nom}}{m} \quad (1)$$

The research team used a virtual refrigerant charge sensor approach presented in [23] since this approach is easier to implement for field-study. This approach requires data acquired from surface-mounted temperature sensors and uses Eq. (2) to calculate FI_{ch} where T_{sc} is measured system subcooling temperature, T_{sh} is measured system superheat temperature, and $T_{sc,rated}$ and $T_{sh,rated}$ are the system superheat and subcooling at standard conditions and values from [7].

$$FI_{ch} = \frac{1}{K_{ch}} \left\{ (T_{sc} - T_{sc,rated}) - \frac{K_{sh}}{K_{sc}} (T_{sh} - T_{sh,rated}) \right\} \quad (2)$$

Although the measured superheat and subcooling values can be calculated using temperature measurements, a more accurate method is to use data acquired from the pressure transducers to calculate the temperatures at given pressures using refrigerant-specific saturation tables. The superheat and subcooling values can then be calculated with Eqs. (3) and (4) where T_{liq} and T_{suc} are temperature of liquid and suction lines respectively, T_{cond} is condensing temperature at the liquid pressure, and T_{evap} is evaporating temperature at the suction pressure [7].

$$T_{sc} = T_{cond} - T_{liq} \quad (3)$$

$$T_{sh} = T_{suc} - T_{evap} \quad (4)$$

Parameters K_{ch} , K_{sh} , K_{sc} in Eq. (2) are calculated using Eqs. (5) and (6) presented in [23] at given subcooling and superheat at rated and operating conditions where α_0 is the ratio of refrigerant charge necessary to have saturated liquid at the exit of the condenser to the rated refrigerant charge and $X_{hs,rated}$ is the ratio of high-side charge to the total refrigerant charge at the rated condition. Li and Braun [23] and Kim and Braun [24] use default values for α_0 and $X_{hs,rated}$; therefore, we use 0.75 for α_0 and 0.73 for $X_{hs,rated}$. FI_{ch} is then compared to a threshold value, to determine whether the systems is overcharged, or undercharged.

$$K_{ch} = \frac{T_{sc,rated}}{(1-\alpha_0)X_{hs,rated}} \quad (5)$$

$$\frac{K_{sh}}{K_{sc}} = \frac{T_{sc}-T_{sc,rated}}{T_{sh}-T_{sh,rated}} \quad (6)$$

Mehrabi and Yuill [17] developed a general relation to calculate the fault impact ratio using Eq. (7) for the refrigerant charge fault (FIR_{ch}) based on the COP of the RTU and the analysis of the experimental data collected from previous studies for A and B standard test conditions for cooling mode based on the AHRI standard 210/410 (2008) [25]. This relationship holds for a single speed fan and compressor system where a_0 , a_1 , and a_2 are regression coefficients. These coefficients depend upon operating conditions and whether the system has a fixed orifice (FXO) or thermostatic expansion valves (TXVs). Here we assumed the A test condition and adapted the coefficient from [17] as

follows: $a_0 = 0.97807$; $a_1 = 0.32443$; $a_2 = -1.47617$ for systems with fixed office expansion valves (FXO) and $a_0 = 1.00920$; $a_1 = 0.17685$; $a_2 = -1.60799$ for systems with thermal expansion valves (TXV).

$$FIR_{ch,COP} = a_0 + a_1 \cdot FI_{ch} + a_2 \cdot FI_{ch}^2 \quad (7)$$

Condenser fouling: Condenser fouling reduces the airflow, which reduces heat rejection from condenser coils to the surroundings. The fault intensity of condenser fouling (FI_{CA}) can be calculated with Eq. (8) where $\dot{V}_{con,nom}$ is the nominal condenser air flow rate specified in the manufacturer's literature and $\dot{V}_{con,act}$ is the actual condenser air flow rate [12].

$$FI_{CA} = \frac{\dot{V}_{con,act} - \dot{V}_{con,nom}}{\dot{V}_{con,nom}} \quad (8)$$

The actual condenser air flow rate in ft^3/sec is calculated using Eq. (9) where

$\dot{m}_{ref}$ is the refrigerant mass flow rate, h_{dis} is the enthalpy at the compressor discharge in Btu/lb at T_{dis} and P_{dis} and h_{liq} are the liquid line enthalpy in Btu/lb at T_{liq} , and T_{CAO} T_{CAI} are air temperatures at the outlet and inlet of the condenser [7].

$$\dot{V}_{con,actual} = \frac{\dot{m}_{ref}(h_{dis} - h_{liq})}{(T_{CAO} - T_{CAI})} \frac{\dot{v}_{con,air}}{1.013} \quad (9)$$

$\dot{m}_{ref}$ is calculated using Eq. (10) when the subcooling temperature is not zero by applying the energy balance on the compressor where $\dot{m}_{ref}$ is the refrigerant mass flow rate, h_{dis} is the enthalpy at the compressor discharge in Btu/lb at T_{dis} and P_{dis} [7]. h_{liq} is the liquid line enthalpy in Btu/lb at T_{liq} , T_{CAO} and T_{CAI} are air temperatures at

the outlet and inlet of the condenser. W_{comp} is the compressor input power, and h_{suc} is the compressor suction enthalpy in Btu/lb at T_{suc} .

$$\dot{m}_{ref} = \frac{0.95.W_{comp}}{h_{dis}-h_{suc}} \quad (10)$$

$\dot{v}_{con,air}$ is the specific volume of air in ft^3/lb and can be calculated by using the measured temperature at the air inlet and outlet to the condenser with Eq. (11).

$$\dot{v}_{con,air} = 0.04512 \left(\frac{(T_{CAI}-T_{CAO})}{2} + 273.15 \right) \quad (11)$$

FI_{CA} is determined to identify a condenser fouling fault by comparing it to a preset threshold. If $FI_{CA} > \text{threshold}$, then condenser fouling is present. Then the general relationship developed in [17] can be used to calculate the fault impact ratio with Eq. (12) for the condenser fouling fault (FIR_{CA}) based on the COP of the RTU where a_0 , a_1 , and a_2 are regression coefficients. Here we assumed the A test condition and adapted the coefficient from [17]: $a_0 = 1.00$; $a_1 = 0.31372$; $a_2 = -0.3647$ for systems with FXO and $a_0 = 1.0093$; $a_1 = 0.79837$; $a_2 = 0.54101$ for systems with a TXV.

$$FIR_{CA,COP} = a_0 + a_1.FI_{CA} + a_2.FI_{CA}^2 \quad (12)$$

Evaporator fouling: Evaporator fouling also reduces airflow and usually occurs due to filter fouling. The fault intensity of evaporator fouling (FI_{EV}) is calculated with Eq. (13) where $\dot{V}_{sup,nom}$ is nominal evaporator air flow rate specified in the manufacturer's literature and $\dot{V}_{sup,act}$ is actual (measured) supply air flow rate over the evaporator coil in ft^3/sec [12].

$$FI_{EV} = \frac{\dot{V}_{sup,act} - \dot{V}_{sup,nom}}{\dot{V}_{con,nom}} \quad (13)$$

FI_{EV} is used to identify an evaporator fouling fault by comparing it to a preset threshold. If $FI_{EV} > \text{threshold}$, then evaporator fouling is present. The general relationship is then developed in [17] to calculate the fault impact ratio for the evaporator fouling fault (FIR_{EV}) based upon COP of the RTU where a_0 , a_1 , and a_2 are regression coefficients using Eq. (14). The A test condition is assumed and so use the coefficients from [17]: $a_0 = 0.99381$; $a_1 = 0.13673$; $a_2 = 0.09088$ for systems with FXO and $a_0 = 0.99322$; $a_1 = -0.02259$; $a_2 = -0.34522$ for systems with TXV.

$$FIR_{EV,COP} = a_0 + a_1 \cdot FI_{EV} + a_2 \cdot FI_{EV}^2 \quad (14)$$

Energy Performance Degradation: Energy degradation can give an overall indication of how well the system is performing. This can be quantified by the COP using Eq. (15), which is the ratio between desired heat removal from the air by the evaporator to the required power input to the compressor, and the EER of the refrigerant side [7,21] where Q_{ref} is the system capacity calculated with Eq. (16), W_{main} is the total system power input, and W_{fan} is the power input to the evaporator fan. $\dot{m}_{ref}$ is calculated from Eq. (10) and takes into consideration the heat loss through the compressor, where $h_{evap,outlet}$ and $h_{evap,inlet}$ are the enthalpies at the evaporator outlet and inlet respectively.

$$COP_{ref} = \frac{Q_{ref} - W_{fan}}{W_{main}} \quad (15)$$

$$Q_{ref} = \dot{m}_{ref}(h_{evap,outlet} - h_{evap,inlet}) \quad (16)$$

Air side EER is also calculated based on the ASHRAE handbook of Fundamentals [21] (details of this calculation is not presented here). The estimated EER value for both the refrigerant side and air side are calculated with Eq. (17) based upon the calculated COP.

$$EER = 3.412 \text{ COP} \quad (17)$$

Economizer Faults: It is difficult to identify economizer faults without knowing the status of the outdoor air damper. Economizer faults can include an air temperature sensor failure, not economizing when it should, economizing when it should not, a damper not modulating, and intake of excess outdoor air. However, the methodology developed in [7] relates the damper position to outdoor air percentage (OA%) using Eq. (18). If OA% is 0% then the damper is fully closed and 100% is fully opened. This method uses the outdoor air temperature (T_{OA}), mixed air temperature (T_{MA}), and return air temperature (T_{RET}) to find the portion of air entering the building.

$$OA\% = \frac{T_{RET} - T_{MA}}{T_{RET} - T_{OA}} \quad (18)$$

Control Faults: Control faults are generally difficult to directly measure and consist of incorrect set points, scheduling errors, or unusual control board malfunctions. Instead of causing wear and degradation to the equipment, these faults increase the energy expenditure through a multitude of faulty or suboptimal operations. To determine inefficient and faulty operations, a power analysis is conducted by monitoring the energy use of the RTU through a main meter and sub meters for the compressor and evaporator fan. Using this energy data an operator is able to determine whether the

unit is active during unoccupied building times, running for either the entire day or for a long duration time when it should not be or short cycling periodically. This power analysis method is demonstrated for Tool 1 in Sec. 6.1.

3.5. Measurement and Verification of Energy Savings

Energy savings due to fault correction will be calculated using Eq. (19).

$$\text{Savings (kWh)} = \text{Faulted energy use (kWh)} - \text{Unfaulted (Retrofit) energy use (kWh)} \quad (19)$$

The annual energy cost savings will be calculated using Eq. (20) by multiplying the energy usage by the cost of electricity (e.g. \$0.12).

$$\begin{aligned} \text{Annual cost savings} \\ = \text{energy savings} * \text{rate} \\ + \sum \text{month}(\text{monthly demand savings} * \text{rate}) \end{aligned} \quad (20)$$

To calculate energy savings, the energy use data, which was obtained by the IMS for faulted and un-faulted (retrofit) scenarios, was correlated with the outdoor ambient temperature or cooling degree-days (CDD) or heating degree-days (HDD) to calculate weather-normalized annual energy savings by FDD tool by site.

4. MARKET EVALUATION

The purpose of the market evaluation is to determine the current market conditions for streamlining adoption of FDD technologies and to analyze stakeholder feedback for the purposes of designing a utility marketing and incentive program that increases adoption

of FDD technologies. The goal is to propose a marketing and incentive program that the study partners or other utility companies can replicate in other states. To identify energy-efficiency technology adoption attitudes and FDD market barriers, two preliminary surveys were conducted at a utility customer outreach event in Connecticut in 2018. Thirty people responded to different aspects of the survey and their roles are characterized in Fig. 3.

Insert Fig. 3 Here

Table 5 shows survey questions developed to understand the perspectives on the value proposition for energy efficiency and participant responses. A five-point Likert scale was applied to understand relative value propositions. Each respondent was asked, “How important are the following value propositions to your organization, customers, or stakeholders when investing in building energy efficiency technologies?” Table 5 depicts the responses from 28 people and calculates the mean and rank order. The results indicate that (1) energy consumption, (2) energy cost, (3) energy demand, and (4) occupant comfort ranked with the highest value priorities followed by (5) operation cost and (6) maintenance cost.

Insert Table 5 Here

Table 6 shows survey questions developed to understand the severity of market barriers to the adoption of FDD technologies. A five-point Likert scale was applied to understand relative market barriers. Table 6 also tabulates the 30 participant responses and calculates the mean and barrier rank order.

Insert Table 6 Here

Figure 4 graphically depicts the survey results for information barriers, organizational barriers, and technical barriers.

Insert Fig. 4 Here

Based on the preliminary survey at the outreach event and previous work [14,26–32], detailed market surveys targeted for different stakeholders (building owners/facility managers; HVAC contractors/energy consultants; utilities; FDD vendors) have been developed and executed with stakeholders participating in the current study. The questions are organized into sections as shown in the appendix: I) Knowledge for FDD Products, II) Awareness for FDD Products, III) Attitudes for FDD products, and IV) Respondent characteristics. The results of the market study will (1) help vendors in developing more effective FDD products, (2) support state Conservation and Load Management (CLM) programs in determining proper incentive and rebates for FDD products, and (3) assist FDD vendors and CLM programs in the development of effective FDD marketing campaigns.

5. PROCESS EVALUATION

The purpose of the process evaluation is to analyze the current processes of purchasing, installing, tuning, and using current FDD tools for retro commissioning (RCM). Common process steps across all installations are shown in the appendix. The process for BMS-based FDD tools is different than equivalent processes for non-BMS, hardware-based

FDD tools. For example, Tool 1, which is a hardware-based tool, was installed at Site 1. A trained technician was able to install it in less than 4 hours. Another hardware-based FDD tool, Tool 2, was installed at Site 2 and this installation proved to be a difficult installation due to limited access to ductwork in the attic space and limited space within the RTU to place sensors, especially the refrigerant pressure sensors. The installation was completed over 1.5 days, including troubleshooting time. The detailed process evaluation of FDD products of the ongoing will be completed during fall 2020. The full process evaluation results and analysis will show how each step differs by FDD tool, will analyze variability by process step across tools, and will offer recommendations to improve these processes for future FDD product installations.

6. RESULTS AND DISCUSSION

The IMS was installed at ten sites in Connecticut to analyze the performance of each FDD product in the field. Data was collected to verify all of the sensors were connected and working properly. Additional visits to the sites are made to rectify monitoring sensor errors when irregularities in sensor data are identified. Preliminary faults detected by the FDD tools are categorized into sensor, control, hardware, and other fault groups and summarized in Table 7 for each tool's baseline period. The IMS data was analyzed and the faults were identified based on the FI and FIR-based methods described in Sec. 3.4.

Insert Table 7 Here

A summary of findings based upon analysis of IMS data are presented in Table 8. The EER is calculated using three different methods; two by using air side performance and one-way by using refrigerant-side performance. This method is applied for redundancy since airflow measurement in the field may not be accurate in some instances. Also, sometimes inaccurate temperature measurements may result in inaccurate refrigerant-side EER. The Main EER is calculated based upon return air and mixed temperatures and represents the performance of cooling, based on the ability to cool the building. The refrigerant-side EER represents the performance of the vapor compression system and uses the refrigerant line temperatures and refrigerant pressures for calculation.

This paper demonstrates the methods employed in the study and discusses only preliminary results for the FDD Tool 1 at Site 1 in detail in Secs. 6.1 and 6.2. Analysis and results for the other nine tool-site pairs are planned as part of future work. Figure 5 displays photos of the installation of the data logger and pressure/temperature sensors for the IMS, while Fig. 6 presents the photos of the FDD Tool 1, which is a hybrid, hardware-based tool.

Insert Table 8 Here

Insert Fig. 5 Here

Insert Fig. 6 Here

6.1. Monitoring Tool Findings

In this paper, Site 1 was selected for presenting detailed results from the independent monitoring tool. Figure 7 presents the calculated overall system EER (air side) for a hot summer day (July 21, 2019). The weather data corresponding to this day for this site is shown in Fig. 8. The rated EER for this RTU is 11, and is denoted in Fig. 7 by the red line. As shown in Fig. 7, the overall EER is slightly over 11.0, indicating no reduction in its performance related to the rated EER.

Insert Fig. 7 Here

Insert Fig. 8 Here

Figure 9 shows percentage of outdoor air entering the building as well as return, outdoor, and mixed air temperatures. As seen in Fig. 9, when the outdoor temperature is low, the unit is bringing in 100% outdoor air and when the outdoor temperature begins to increase, the outdoor air is reduced to about 20% indicating minimum ventilation air. This result shows that the economizer seems to be working as desired. By examining the outdoor, return, and mixed air temperatures, one can glean how much outdoor air draws through the economizer.

Insert Fig. 9 Here

Figure 10 represents the fault intensity (FI) of the refrigerant charge for both refrigerant circuits. The FI value indicates a slight overcharge for circuit A (about +6%) and a slight overcharge for Circuit B (about +15%). It is known that refrigerant overcharge has little effect on EER and the only concern, generally, is that it may flood the compressor and

damage it. Since there is no significant reduction in EER based upon the calculations, overcharge is not a large concern. Since the RTU at this site has TXV, subcooling is calculated and used to verify the refrigerant charge impact. Subcooling temperatures are illustrated in Fig. 11. For circuit A, the subcooling is in the range of about 12-14°C and for circuit B, it is in the range of 16-18°C. Typical subcooling for TXV-based systems is in the range of about 8-10°C indicating both circuits are slightly overcharged.

Insert Fig. 10 Here

Insert Fig. 11 Here

The fault intensity (FI) for evaporator and condenser fouling faults were also calculated and found to be acceptable, indicating no fouling faults.

Finally, the RTU runtime was examined to see if there is excessive cycling of the unit. Figure 12 presents the power for the main and its components. As shown in Fig. 12, there is cycling of the unit during nighttime, but the system is on continuously (at least one compressor stage) on during the day. However, this July day was a hot summer day. During nighttime, there were 3 cycles per hour, indicating the system has been sized properly and that the cycling losses are minimal.

Insert Fig. 12 Here

6.2. Analysis of Results and Discussion

As presented above, FDD Tool 1 is a hybrid tool. A sample screenshot of the FDD Tool 1 dashboard is shown in Fig. 13. It shows various monitored variables as well as whether

there are any issues related to energy performance, health of the unit, and comfort.

The color green indicates there are no issues. For the initial baseline monitoring period, the FDD Tool did not indicate any performance, health, or comfort faults for the RTU at Site 1. The Tool indicated only a few transient drive fault alarms, related to its variable frequency drive (VFD), and they resolved themselves as shown in Fig. 14. These alarms did not have any impact on the RTU performance and the FDD tool indicated that the RTU at Site 1 was performing well during the initial baseline period. The IMS tool did not identify any faults for this site as discussed in Sec. 6.1. However, this limited verification of FDD Tool 1 does not indicate how effective it is in identifying faults. Unfortunately, it is the reality when demonstrating such tools in the field operating conditions.

Insert Fig. 13 Here

Insert Fig. 14 Here

Also, the capability of IMS in identifying faults can be verified by introducing controlled faults artificially. However, this approach would interfere with study that is based on naturally occurring faults. Also, getting the approval from the building owners to introduce faults into RTUs is a barrier for recruiting sites for study participation.

7. CONCLUSIONS AND FUTURE WORK

The importance of FDD tools and the need for a comprehensive approach to bring the FDD tools into the mainstream are highlighted in this work. A methodology for

evaluating the technical performance was proposed for FDD tools installed on RTUs operating under real conditions. The methodology was demonstrated for one installed FDD tool. The baseline testing will continue through summer of 2021 and recommendations for repairs and retro commissioning of RTUs will be identified by the data analysis using the IMS and FDD tools. Once the retro commissioning is complete, the RTUs will be monitored through continuous commissioning and energy savings potential will be calculated. Also, the capability of IMS in identifying faults can be verified by introducing controlled faults artificially. This approach would interfere with study that is based on naturally occurring faults. However, this task could be attempted at the end of the project. This analysis, results, and recommendations for all sites will be provided in future work. This research also develops and defines an approach for conducting market and process evaluations for FDD products in the field. Detailed results and analysis of the market and process evaluations will also be provided in future work.

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The authors would like to thank the United Illuminating Company and Eversource employees for helping the research team to recruit sites for the study and for review of proposed methods and results. We would also like to thank building owners, energy managers, mechanical contractors, and FDD vendors for participating in this study.

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NOMENCLATURE

BAS =	building automation and control system
BMS =	building management system
COP =	coefficient of performance
EER =	energy efficiency ratio
EMS =	energy management system
FDD =	fault detection and diagnostics
FI =	fault intensity
FIR =	fault impact ratio
FXO =	fixed office expansion valve
h =	enthalpy
IMS =	independent monitoring system
m =	mass
$\dot{m}$ =	mass flow rate
P =	pressure
Q =	system capacity
RTU =	rooftop unit
T =	temperature
TXV =	thermal expansion valve
$\dot{V}$ =	air flow rate
$\dot{v}$ =	specific volume of air
W =	power
α_0 =	ratio of refrigerant charge necessary to have saturated liquid at the exit of the condenser to the rated refrigerant charge
$X_{hs, rated}$ =	ratio of high-side charge to the total refrigerant charge at the rated condition

Subscripts

act = actual

CA = condenser fouling

CAI = condenser air inlet

CAO = condenser air outlet

ch = refrigerant charge

comp = compressor

cond = condenser

dis = discharge

EV = evaporator fouling

evap = evaporator

liq = Liquid line

MA = mixed air

nom = nominal

OA = outdoor air

ref = refrigerant

RET = Return air

sc= subcooling

sh = superheat

suc = Suction line

sup = supply

APPENDIX

Table 9 provides more details on the specification of the FDD evaluator tool and the sensors used as part of its sensor network. The first column describes the sensor, the second column describes the model number and the third column provides further details and specifications of the sensor. All sensors were pretested in the laboratory before installing in the field evaluator tool.

Insert Table 9 Here.

The goal of the market study is to determine the levels of knowledge, awareness, and attitudes that exist for study participants that may influence the adoption of FDD technology. The market study will be conducted with all study participants including customers, HVAC and controls technicians, FDD vendors, and utility program managers and engineers. Four market studies were reviewed to determine best practices for conducting this market study and survey: (1) 2012 Connecticut Efficient Lighting Saturation and Market Assessment [32], (2) 2015 Barriers to Commercial and Industrial Program Participation with a Focus on Financing and Cancellations [26], (3) 2014 Connecticut Ground Source Heat Pump Impact Evaluation & Market Assessment [28], and (4) 2015 Connecticut Commercial & Industrial (C&I) Market Research [31]. Table 10 depicts the survey questions grouped into four sections: (1) customer knowledge for FDD products, (2) customer awareness for FDD Products, (3) customer attitudes for FDD products, and (4) respondent characteristics. Descriptive statistics will be calculated and

reported for each level of knowledge, awareness, and attitude for FDD adoption. Advanced analysis will be conducted to determine if correlations exist between knowledge, awareness, and attitudes and the respondent's intention to adopt FDD technology. Results of this study will be used to design marketing, outreach, and education programs that educate customers on the use of FDD technologies.

Insert Table 10 Here.

The goal of the process study is to record the current process for installation of FDD products in the field. The research team monitored major installation steps and recorded what went well and where there were issues or problems with an FDD installation. Table 11 indicates 23 major steps that occurred in some or all of the installations. Typical steps, times, and issues will be reported in the process study as well as variance between installations.

Insert Table 11 Here.

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Figure Captions List

- Fig. 1 A schematic of Intellilogger, a data acquisition and monitoring system, used as a verification system (IMS evaluator tool)
- Fig. 2 A schematic of a rooftop HVAC unit with labels indicating sensor locations for the IMS evaluator tool
- Fig. 3 Outreach event preliminary market survey participant characteristics
- Fig. 4 Analysis and results of outreach event preliminary market survey
- Fig. 5 The installation of the Independent Monitoring System (IMS) at site 1
- Fig. 6 Photos of the installed FDD Tool 1 at site 1
- Fig. 7 Calculated air side EER based upon the measurement data from IMS evaluator tool for Site 1
- Fig. 8 Outdoor air temperature and relative humidity profiles for Site 1 on July 21, 2019
- Fig. 9 Percentage of outdoor air entering the building; return, outdoor, and mixed air temperatures of the RTU at Site 1 from IMS evaluator tool
- Fig. 10 Fault Intensity of refrigerant charge for the RTU at Site 1 from IMS evaluator tool
- Fig. 11 Subcooling for refrigerant circuits A and B of RTU at Site 1 from IMS evaluator tool
- Fig. 12 RTU runtime with measured powers for various components of the system
- Fig. 13 A sample screenshot of FDD Tool 1's dashboard
- Fig. 14 A screenshot of Tool 1 platform showing the detected VFD drive faults

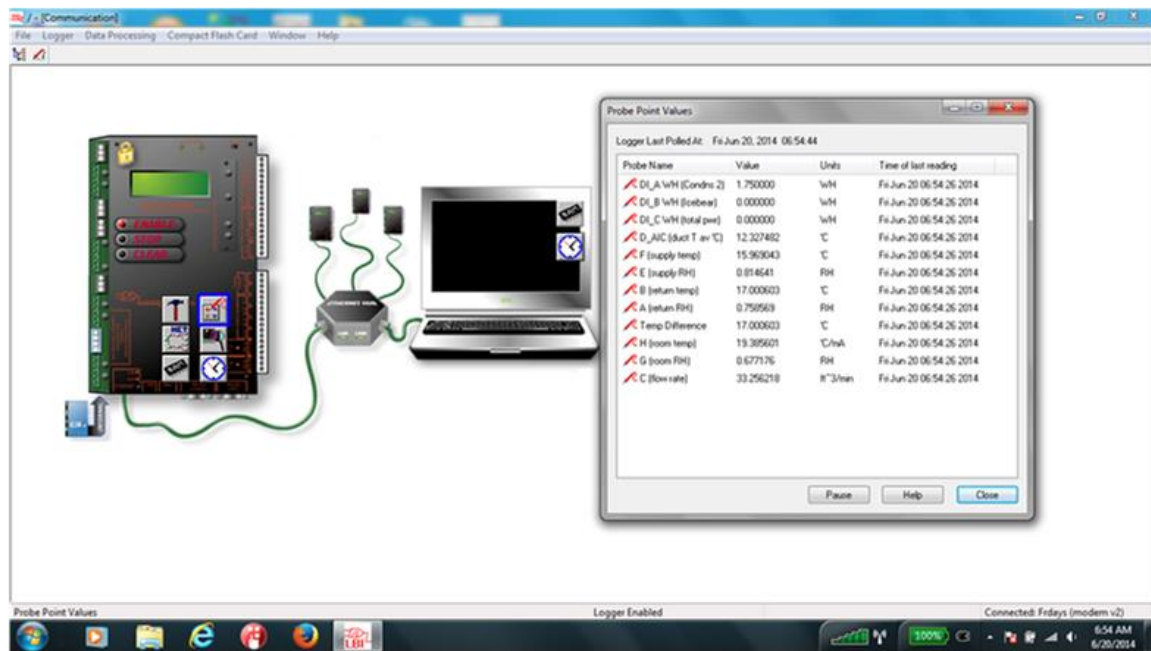


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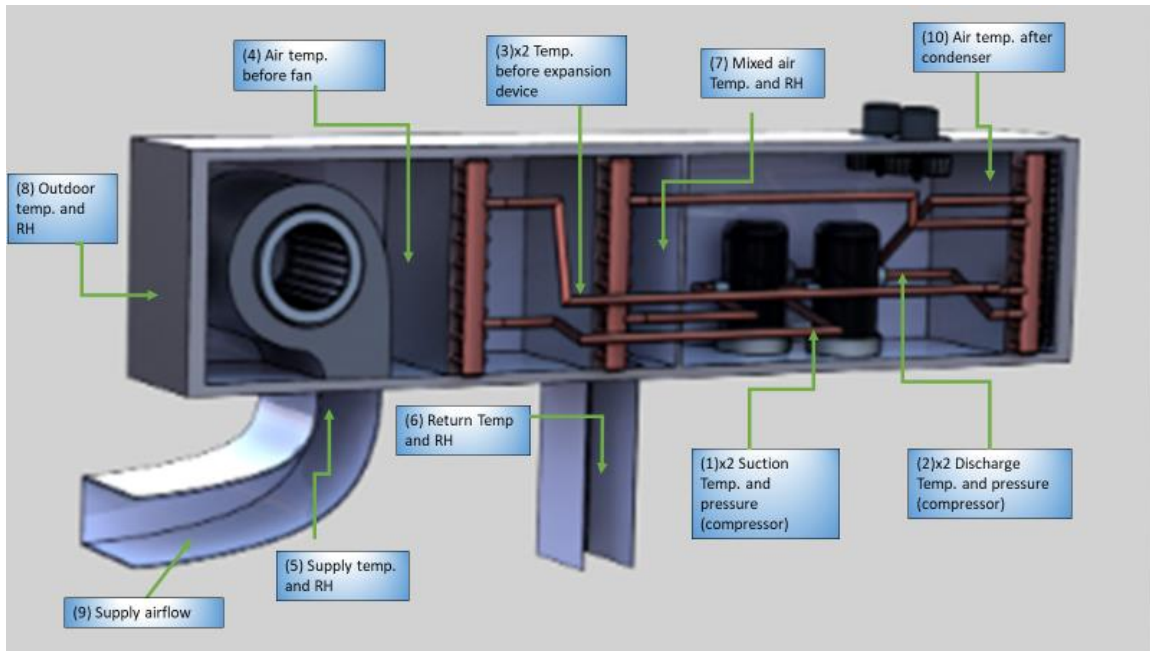


Fig. 2 A schematic of a rooftop HVAC unit with labels indicating sensor locations for the IMS evaluator tool

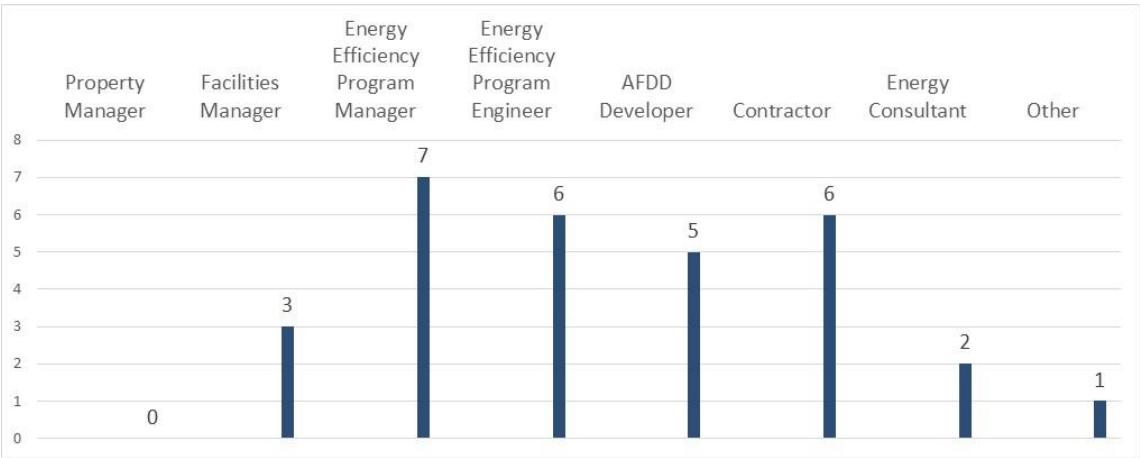


Fig. 3 Outreach event preliminary market survey participant characteristics

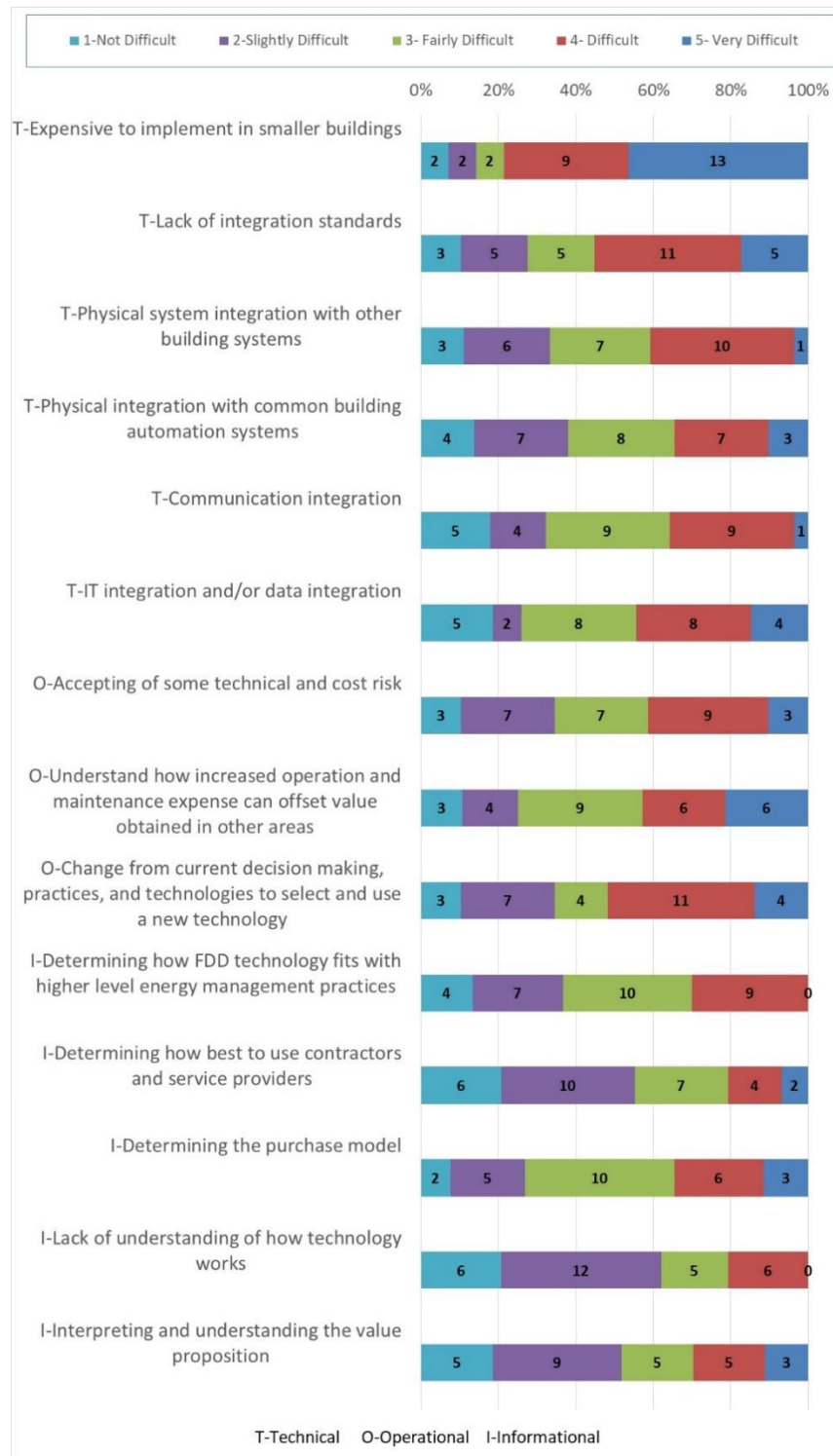


Fig. 4 Analysis and results of outreach event preliminary market survey



Fig. 5 The installation of the Independent Monitoring System (IMS) at site 1



Fig. 6 Photos of the installed FDD Tool 1 at site 1

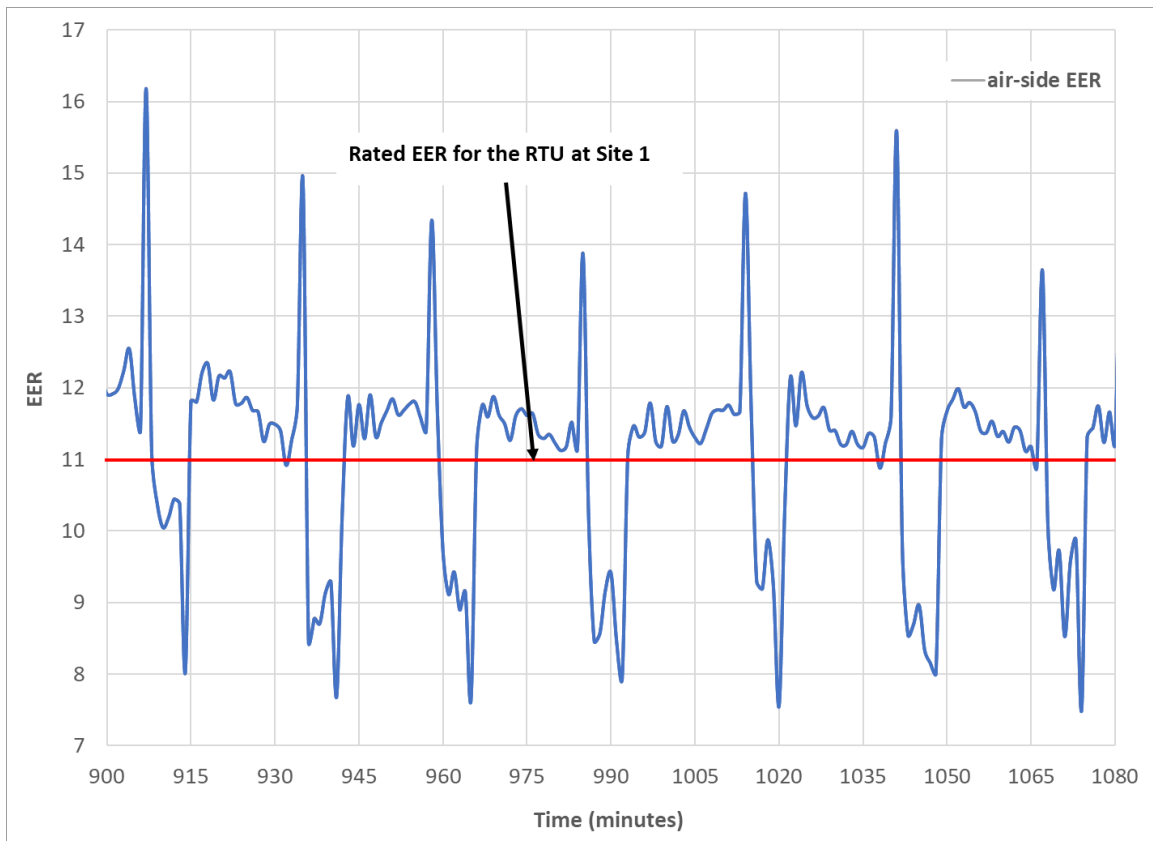


Fig. 7 Calculated air side EER based upon the measurement data from IMS evaluator tool for Site 1

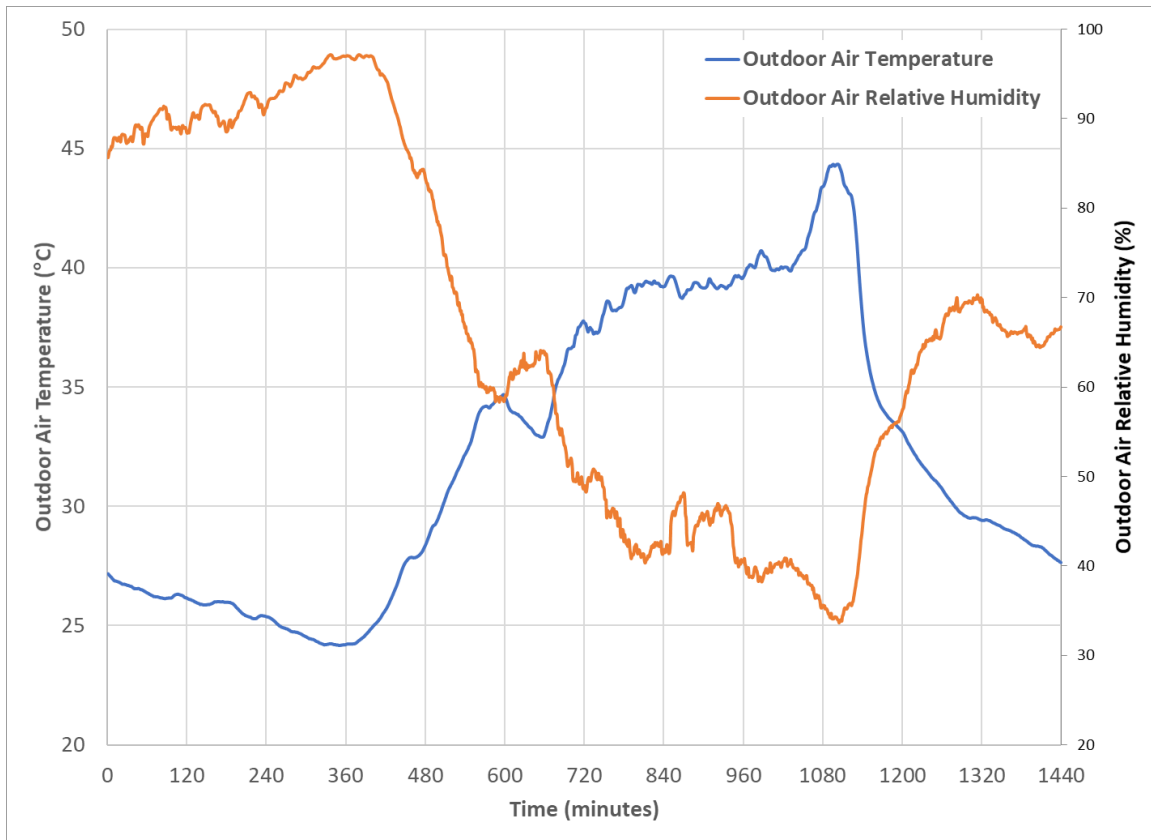


Fig. 8 Outdoor air temperature and relative humidity profiles for Site 1 on July 21, 2019

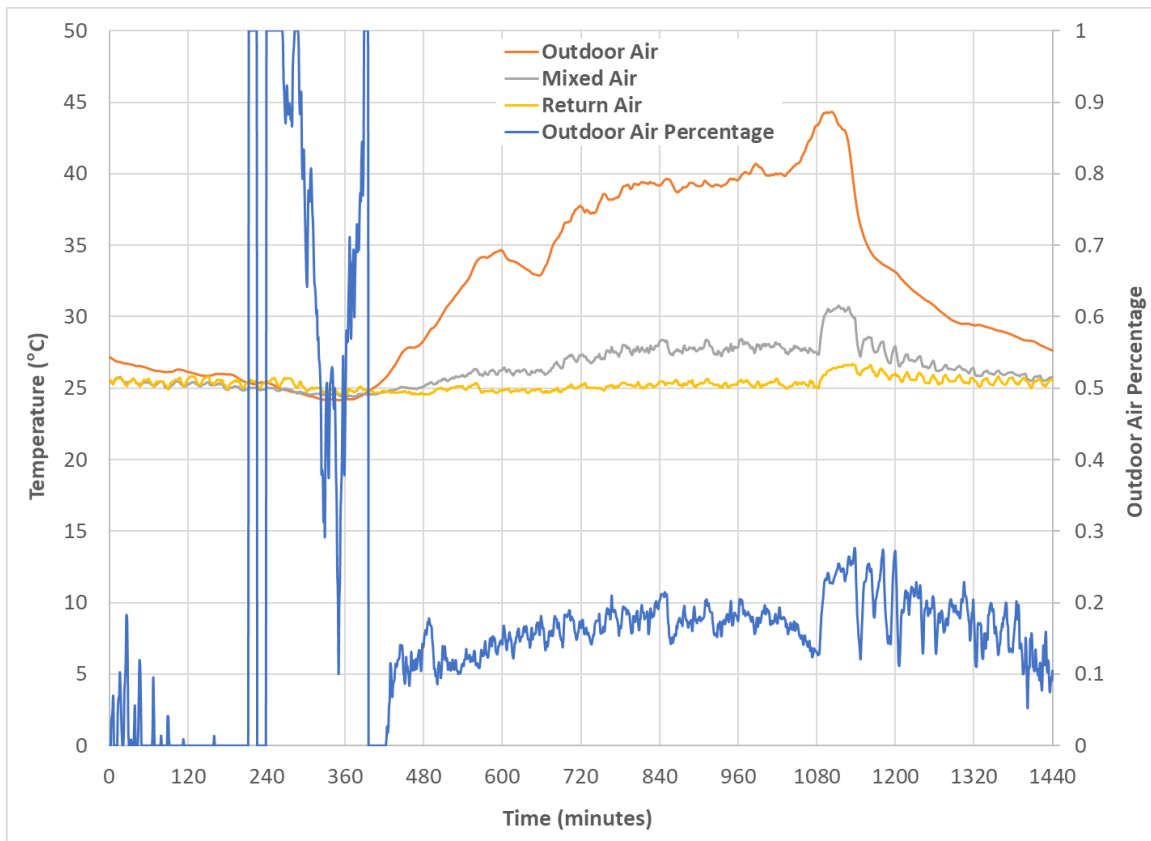


Fig. 9 Percentage of outdoor air entering the building; return, outdoor, and mixed air temperatures of the RTU at Site 1 from IMS evaluator tool

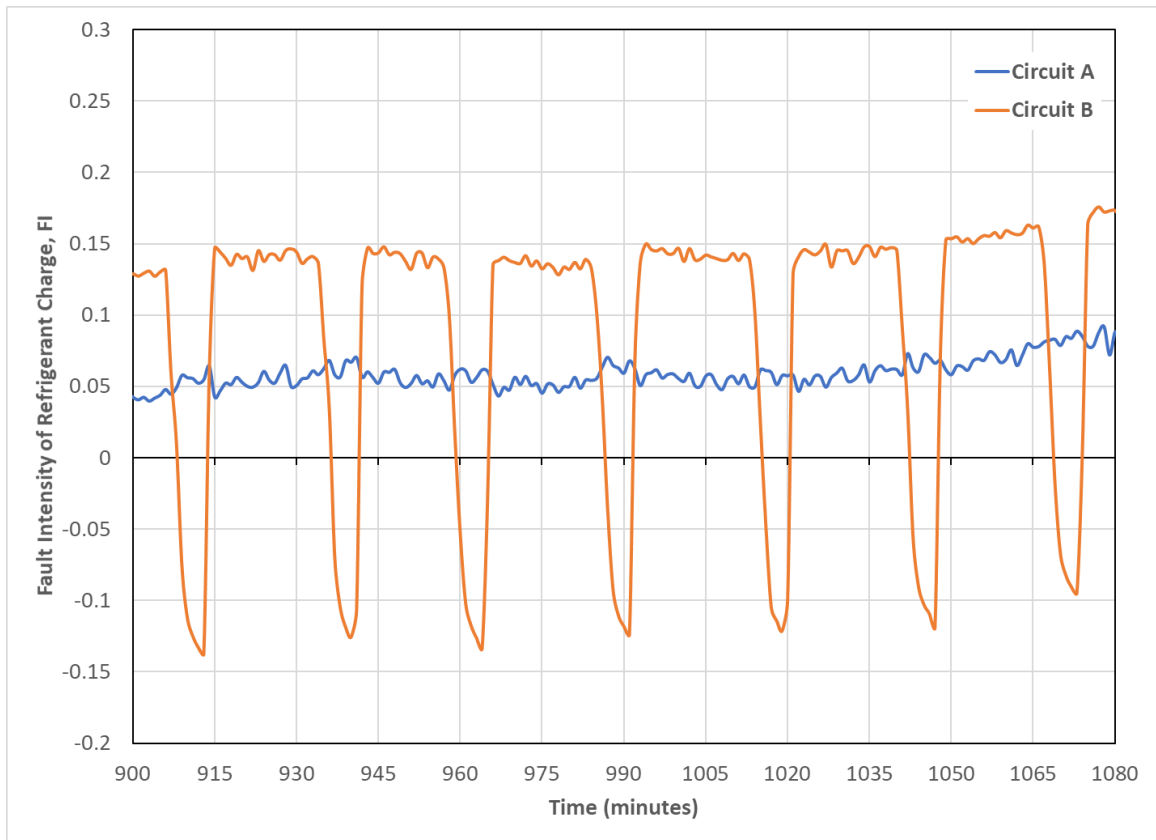


Fig. 10 Fault Intensity of refrigerant charge for the RTU at Site 1 from IMS evaluator tool

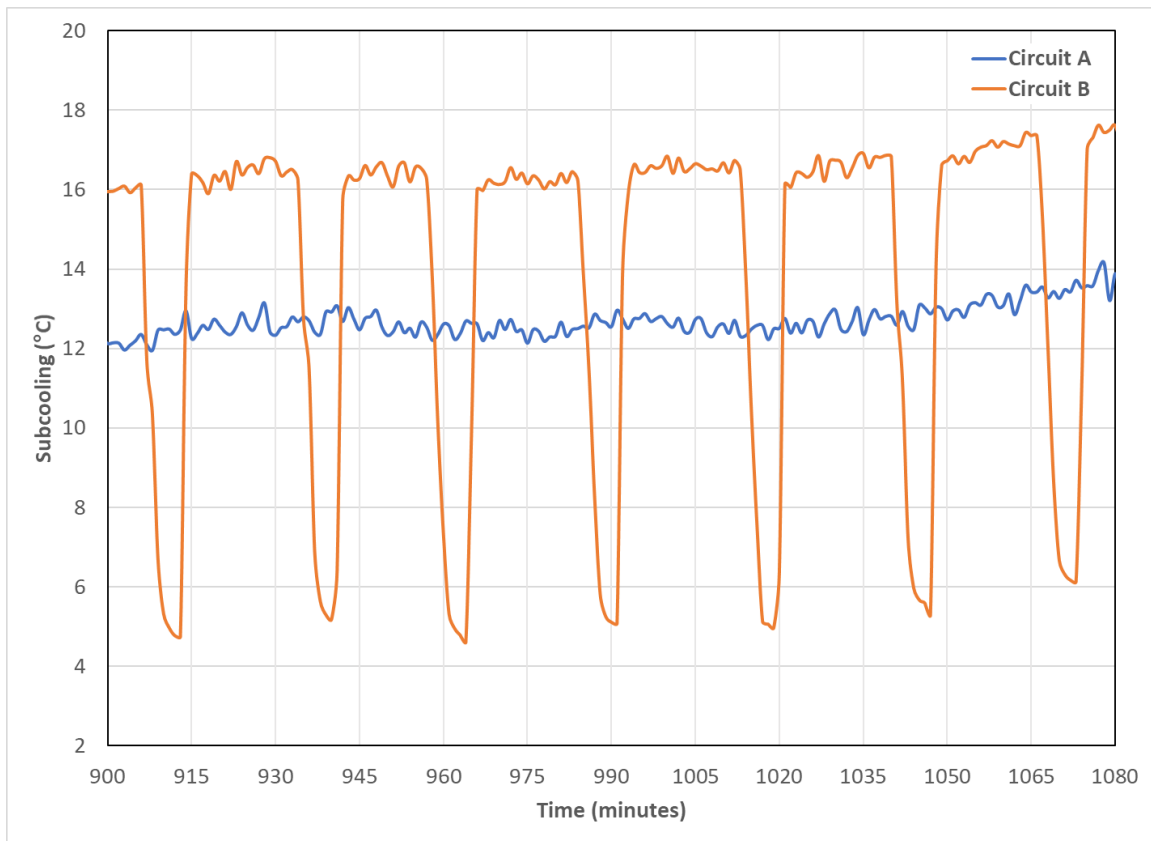


Fig. 11 Subcooling for refrigerant circuits A and B of RTU at Site 1 from IMS evaluator tool

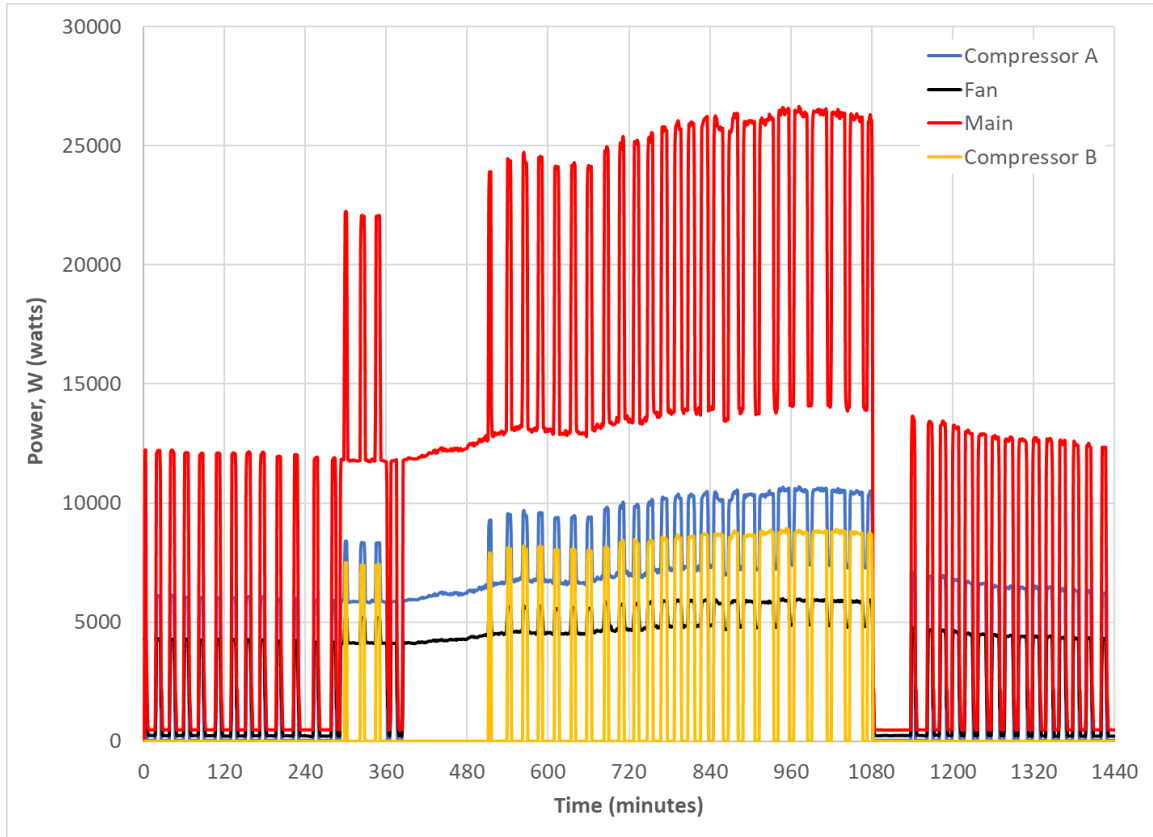


Fig. 12 RTU runtime with measured powers for various components of the system



Fig. 13 A sample screenshot of FDD Tool 1's dashboard

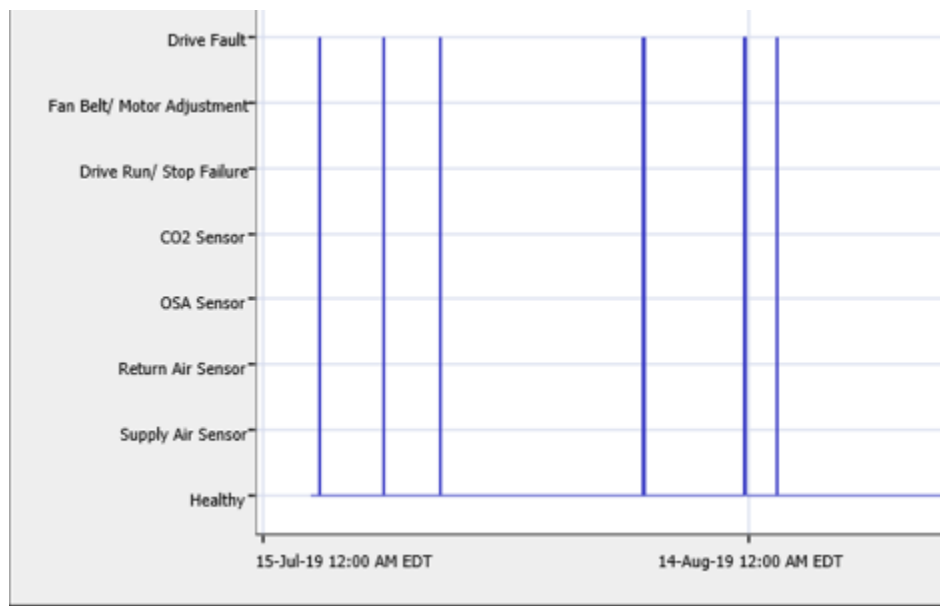


Fig. 14 A screenshot of Tool 1 platform showing the detected VFD drive faults

Table Caption List

Table 1	Overall normalized scoring of FDD tools based upon vendor RFI responses
Table 2	Site selection factors considered in the study
Table 3	List of sites for FDD tool installation
Table 4	Sensors and instrumentation used for IMS
Table 5	A preliminary market survey on the value of energy efficiency objectives in buildings
Table 6	A preliminary survey of FDD market barriers with results
Table 7	Summary of faults identified by FDD tools for each site during first baseline period
Table 8	Summary of faults identified for each site by the IMS
Table 9	Specifications for IMS sensors and instrumentation
Table 10	Market survey questionnaire
Table 11	Common process steps for process evaluation methodology

Table 1 Overall normalized scoring of FDD tools based upon vendor RFI responses

FDD tool	Raw score	Percent	Tool type
Reference, max	17.89	100%	NA
Tool 1	12.92	72%	hybrid
Tool 2	16.79	94%	hardware
Tool 3	13.64	76%	hardware
Tool 4	13.37	75%	hardware
Tool 5	16.49	92%	software
Tool 6	16.44	92%	software
Tool 7	14.54	81%	software
Tool 8	11.60	65%	hardware
Tool 9	15.29	85%	hybrid
Tool 10	14.65	82%	software
Tool 11	15.74	88%	hybrid
Tool 12	17.39	97%	hardware

Table 2 Site selection factors considered in the study

Site factor	Factor metric
Building use	Building type
	Year-round use
RTU characteristics	RTU characteristics: size, age, type
RTU and ductwork access	RTU accessibility
	Ductwork accessibility for installing sensors for M&V
Building system characteristics	Building energy management system
	Network access capabilities and security
	Recent and planned energy projects
	Status of current building and operations
	Commercial refrigeration
Maintenance practice	Current RTU service status
	Preventative maintenance schedule
	Previous repairs and maintenance
Potential future saving	Nationwide chain or multiple sites

Table 3 List of sites for FDD tool installation

Site	Building type	FDD tool	Tool type
Site 1	Manufacturing facility	Tool 1	Hardware
Site 2	Restaurant	Tool 2	Hardware
Site 3	University dormitory	Tool 3	Hardware
Site 4	Distribution center	Tool 4	Hardware
Site 5	Municipal office building	Tool 5	software
Site 6	University services	Tool 6	software
Site 7	K-12 school	Tool 7	software
Site 8	Retailer	Tool 8	software
Site 9	Health & racquet club	Tool 9	software
Site 10	Corporate research center	Tool 10	software

Table 4 Sensors and instrumentation used for IMS

Sensor type	Location	Number
Temperature	Suction line	1
	Discharge line	2
	Air side after condenser	3
	Before the expansion device	4
	Air at evaporator outlet before the fan	5
Temperature and relative humidity	Supply duct	6
	Return duct	7
	Mixed air	8
	Outdoor	9
	Indoor	10
	Supply duct averaging	11
Airflow	Return duct averaging	12
	Supply duct	13
Power meter	Compressor, blower, main	14
Current transformers	Each leg of compressor, blower, and main	15
Pressure transducer	Suction line	16
	Discharge line	17

Table 5 A preliminary market survey on the value of energy efficiency objectives in buildings

A preliminary market survey on the value of energy efficiency objectives in buildings								
How important are the following value propositions to your organization, customers, or stakeholders when investing in building energy efficiency technologies?	5-Very important	4-Important	3-Fairly important	2-Slightly important	1-Not important	Do not know	Mean	Rank
Reduce energy consumption	23	4	1	0	0	0	4.79	1
Reduce energy demand	19	4	4	1	0	0	4.46	3
Reduce energy costs	23	3	2	0	0	0	4.75	2
Reduce other utility cost	12	8	5	2	1	0	4	8
Facility operation costs	14	12	2	0	0	0	4.43	5
Facility maintenance costs	16	9	2	0	0	1	4.36	6
Operational performance: customer comfort levels for HVAC-R or product quality for refrigeration	16	9	3	0	0	0	4.46	4
Operational performance: cooling quality(product	13	11	3	1	0	0	4.29	7

Table 6 A preliminary survey of FDD market barriers with results

Fault detection and diagnostics project stakeholder survey									
How difficult do you precise each barrier you or your organization might face in adapting FDD technologies at you facility or your customers' facilities?		Very important	Important	Fairly important	Slightly important	Not important	Do not know	Mean	Rank
Informational	Interpreting and understanding the value proposition	3	5	5	9	5	3	3.3	11
	Lack of understanding of how technology works	0	6	5	12	6	1	2.83	14
	Determining the purchase model	3	6	10	5	2	4	3.49	7
	Determining how best to use contractors and service providers	2	4	7	10	6	1	3.05	13
	Determining how FDD technology fits with higher level energy management practices	0	9	10	7	4	0	3.17	12
Organizational	Change from current decision making, practices, and technologies to select and use new technology	4	11	4	7	3	1	3.69	4
	Understand how increased operation and maintenance expenses can offset value obtained in other areas	6	6	9	4	3	2	3.76	3
	Accepting of some technical and cost risk	3	9	7	7	3	1	3.52	6
Technical	IT integration and /or data integration	4	8	8	2	5	3	3.68	5
	Communication integration	1	9	9	4	5	2	3.35	10
	Physical integration with common building automation system	3	7	8	7	4	1	3.42	8
	Physical system integration with other building system	1	10	7	6	3	3	3.4	9
	Lack of integration standards	5	11	5	5	3	1	3.8	2
	Expensive to implement in smaller building	13	9	2	2	2	2	4.4	1

Table 7 Summary of faults identified by FDD tools for each site during first baseline period

Site	Tool	Fault type				Comments
		Sensor	Control	Hardware	Other	
Site 1	Tool 1	x	-	-	-	Drive fault
Site 2	Tool 2			x	x	Low subcool; low evaporator efficiency; Low compressor efficiency; economizer fault; Low condenser efficiency; high superheat; high evaporation temperature
Site 3	Tool 3	-	-	-	-	None have been detected
Site 4	Tool 4	-	-	-	-	None have been detected
Site 5	Tool 5	x	x	-	-	indoor air temp set point fault; discharge damper manual override fault
Site 6	Tool 6	x	-	-	-	Occupancy sensor fault
Site 7	Tool 7	x	x	-	x	improper cooling staging; sensor mismatch; supply air too high
Site 8	Tool 8	x	x	x	x	Cooling short cycling; excessive cooling; fan short cycling; Low discharge air temp; OA damper open overnight
Site 9	Tool 9	-	-	-	-	None have been detected
Site 10	Tool 10	-	-	-	-	Not yet started on FDD analytics

Table 8 Summary of faults identified for each site by the IMS

Site	Fault type						Overall health
	EER degradation	Refrigerant undercharge	Refrigerant overcharge	Economizer fault	Condenser fault	Evaporator fault	
Site 1	-	-	x ^{a,b}	-	-	-	Good
Site 2	-	x ^a	-	x	x	-	Average
Site 3	-	-	-	-	x	x	Average
Site 4	x	x ^a	-	-	x	-	Poor
Site 5	-	x ^a	-	x	-	-	Average
Site 6	-	-	-	-	-	x	Good
Site 7	-	-	-	-	-	x	Good
Site 8	x	x ^{a,b}	-	-	x	-	Needs immediate attention
Site 9	-	-	-	x	-	x	Average
Site 10	-	-	x ^{a,b}	-	-	-	Good

^a compressor A; ^b compressor B

Table 9 Specifications for IMS sensors and instrumentation

Sensor/ Variable	Model	Specification
Power meter (main, compressor, fans)	Watt node WNB-3D-400-P0Hz=300	Wattnode pulse output energy meter; 3ph 3-wire 400V (no neutral) or 3ph 4-wire 230V/400V. Custom 300Hz full-scale output frequency (adds 10%)
Supply Duct Avg Temp	Kele ST-FZR85-18-XN3 with T85U-3XW-M	$\pm 0.36^{\circ}\text{F}$ ($\pm 0.2^{\circ}\text{C}$) accuracy, (XN3 NIST certificate, three reference points $32^{\circ}\text{F}/77^{\circ}\text{F}/158^{\circ}\text{F}$ ($0^{\circ}\text{C}/25^{\circ}\text{C}/70^{\circ}\text{C}$))
Return Duct Average Temp	Kele ST-FZR85-18-XN3 with T85U-3XW-M	$\pm 0.36^{\circ}\text{F}$ ($\pm 0.2^{\circ}\text{C}$) accuracy, (XN3 NIST certificate, three reference points $32^{\circ}\text{F}/77^{\circ}\text{F}/158^{\circ}\text{F}$ ($0^{\circ}\text{C}/25^{\circ}\text{C}/70^{\circ}\text{C}$))
Refrigerant Temp (suction, discharge, condenser outlet)	Logic Beach TC-TTU5-0.5-3UT-S-24AWG	0.125"dia x 0.5" 316SS probe with potted ungrounded junction and 20ft of 24AWG lead wire.
Air Temp before fan	Logic Beach TC-TVT6-00-20-S0	20ft 22AWG Teflon stranded leads with Teflon fused tip for low mass thermal response and media protection. Part meets "1/2 Std. Limits" spec resulting in improved accuracy of $\pm 0.3^{\circ}\text{C}$ over 0 to 125C range.
Air Temp after condenser	Logic Beach TC-TVT6-00-20-S0	20ft 22AWG Teflon stranded leads with Teflon fused tip for low mass thermal response and media protection. Part meets "1/2 Std. Limits" spec resulting in improved accuracy of $\pm 0.3^{\circ}\text{C}$ over 0 to 125C range.
Duct Airflow	Flow Measurement Technology NL 1x3	$\pm 2\%$ of reading, 3 nodes, single strut, NIST, Thermal Dispersion Airflow
Return Temp/RH	Setra SRH1-3P-D-11-T3-N	0-99% RH, 3% humidity accuracy, -58 to 140°F temp, 4-20mA out, $\pm 1.1^{\circ}\text{F}$ accuracy

Supply Temp/RH	Setra SRH1-3P-D-11-T3-N	0-99% RH, 3% humidity accuracy, -58 to 140°F temp, 4-20mA out, +/- 1.1°F accuracy
Mixed Temp/RH	Setra SRH1-3P-D-11-T3-N	0-99% RH, 3% humidity accuracy, -58 to 140°F temp, 4-20mA out, +/- 1.1°F accuracy
Outdoor Temp/RH	Setra SRH1-3P-O-11-T3	0-99% RH, 3% humidity accuracy, -58 to 140°F temp, 4-20mA out, +/- 1.1°F accuracy
Indoor Temp and RH	Setra SRH1-3P-W-11-T3	0-99% RH, 3% humidity accuracy, -58 to 140°F temp, 4-20mA out, +/- 1.1°F accuracy
Solar Radiation	LICOR L-Cor 200SZ	Solar Radiation Pyranometer; Includes mounting and leveling base and 50' extension lead.
Refrigerant Pressure Transducer	Dwyer 628CR-75-GH-P2-E3-S1-NIST	1% full scale, 4-20 mA, 1/4" female NPT, 0-10 bar
Refrigerant Pressure Transducer	Dwyer 628CR-81-GH-P2-E3-S1-NIST	1% full scale, 4-20mA, 1/4" female NPT, 0-40 bar
Data logger	Logic Beach IL-80	IntelliLogger IL-80, Data Acquisition and Reporting Instrument, network enabled with integral Web Page server. Includes USB cable, Ethernet cable, 120Vac to 12Vdc power adapter

Table 10 Market survey questionnaire

Section 1: Customer knowledge for fault detection and diagnostics (FDD) products	
1.	How would you characterize your knowledge level or experience with fault detection and diagnostics (FDD) products for HVAC-R systems?
2.	What capabilities and characteristics should a good FDD tool possess?
3.	Rank order the following HVAC-R faults from highest priority to lowest priority that you would like an FDD tool to detect.
4.	Who do you think has more influence to promote and explain the benefits of FDD products in terms of energy efficiency and saving for you?
5.	What information do you look to help you to make a decision when purchasing new products like FDD?
6.	How could mechanical and electrical contractors help you better understand FDD products or explain and promote these products to you?
Section 2: Customer awareness for fault detection and diagnostics (FDD) products	
7.	Is your company aware of the existence of fault detection and diagnostics (FDD) products for HVAC and refrigeration systems? How would you characterize your awareness?
8.	Have you ever asked your mechanical contractor, HVAC controls company, or energy consultant about FDD products?
9.	Has anyone approached you to purchase FDD products? Who are the parties that get involved in these activities?
10.	Do you think utility outreach efforts could help to spread the adoption and use of FDD products?
Section 3: Customer attitudes for fault detection and diagnostics (FDD) products	
11.	What are your key considerations for purchasing and using FDD products? What are you trying to improve or gain? Select all that apply.
12.	Would any of the following motivate you to adopt FDD products, beyond the key considerations you identified in the last question?
13.	Would including an installation and support option from an FDD vendor in the purchase price encourage you to adopt FDD products?

14. Do you own a building that has tenants that pay their own utility bills?
15. Do you think the long-term financing option for adopting FDD will help to increase adoption and use of FDD products?
16. In your opinion what is the most significant activity that utility energy efficiency programs could do to encourage businesses to purchase and install FDD products?
17. Who is the right person in your business to contact for doing an FDD project? Who is the final decision maker for investments in the building?
18. Rank order from highest level of barrier to lowest level these information barriers to the adoption of FDD products.
19. Rank order from highest level of barrier to lowest level these value proposition barriers to the adoption of FDD products.
20. Rank order from highest level of barrier to lowest level these operational barriers to the adoption of FDD products.
21. Rank order from highest level of barrier to lowest level for these technical barriers to the adoption of FDD products
22. How could these barriers or disincentives be removed or minimized?
23. Do you have a negative perception towards FDD products? Are you likely to not investigate or adopt FDD products?

Section 4: Respondent characteristics

24. Which of the following best describes your industry type? Select all that apply.
 25. Which of the following categories best explain your role in FDD products? Select all that apply.
 26. What is your largest building size?
 27. What is your job title?
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Table 11 Common process steps for process evaluation methodology

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1. Select the buildings and units for FDD to be installed
 2. Determine FDD tool selection criteria: hardware-software, others
 3. Find FDD vendors and tools that meet selection criteria
 4. Contact FDD vendors and request quotation/proposal, request demos and case studies
 5. Collect the building and unit information that vendors require and send
 6. Receive quotation/proposal from FDD vendor
 7. Evaluate for information security and obtain permission from IT and facilities to connect and install tool
 8. Order FDD tool
 9. Confirm delivery date
 10. Receive and inspect hardware
 11. Receive installation instructions
 12. Coordinate contractor to install hardware and schedule installation date
 13. Coordinate installation date with people internal to organization
 14. Observe or oversee hardware installation, noting any issues or problems
 15. Resolve any installation, connection, or integration problems
 16. Return and exchange hardware items if needed
 17. Receive notification that installation is complete and working
 18. Request and receive FDD portal login information from vendor or technician
 19. Obtain technical support and request and receive FDD portal use information
 20. Schedule and attend a demonstration or training for portal use
 21. Add, modify, or delete the original rules
 22. Request and receive a report on baseline unit performance or self-create
 23. Plan actions based upon baseline unit performance
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